

$$\begin{aligned}
 1. \Phi_{\text{elec}} &= \int \vec{E} \cdot \hat{n} dA \\
 &= \int E \cos 60 dA \\
 &= \int 300 \cos 60 dA \\
 &= 300 \cos 60 \times A \\
 &= 300 \cos 60 \times \pi (0.02^2) \\
 &= 0.188 \frac{\text{Nm}^2}{\text{C}}
 \end{aligned}$$

2. Only the left and right faces have a non-zero flux.

$$\begin{aligned}
 \Phi_{\text{left}} &= \int \vec{E} \cdot \hat{n} dA \\
 &= \int -E dA \\
 &= -EA = -400(0.02)(0.03) = -0.24 \frac{\text{Nm}^2}{\text{C}}
 \end{aligned}$$

$$\begin{aligned}
 \Phi_{\text{right}} &= \int \vec{E} \cdot \hat{n} dA \\
 &= \int E dA \\
 &= EA = 1000(0.02)(0.03) = 0.6 \frac{\text{Nm}^2}{\text{C}}
 \end{aligned}$$

$$\text{Now } \Phi_{\text{total}} = 0.6 - 0.24 = 0.36 \frac{\text{Nm}^2}{\text{C}}$$

$$\text{Gauss' Law states: } \oint \vec{E} \cdot \hat{n} dA = \frac{\Sigma q_{\text{inside}}}{\epsilon_0}$$

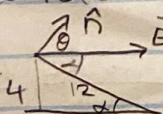
$$\text{So } 0.36 = \frac{\Sigma q_{\text{inside}}}{8.85 \times 10^{-12}}$$

$$\text{So } \Sigma q_{\text{inside}} = 3.19 \times 10^{-12} \text{ C}$$

3. Only the left and right faces have a non-zero flux (the rest have $\vec{E} \cdot \hat{n} = 0$ ($\perp$ to each))

$$\text{So } \oint \vec{E} \cdot \hat{n} dA = \int E_1 (-1) dA + \int E_2 (1) \cos \theta dA$$

Considering the right face:



$$\sin \alpha = \frac{4}{12} \Rightarrow \alpha = 19.5^\circ$$

$$\text{So } 90^\circ = \theta + \alpha \Rightarrow \theta = 70.5^\circ$$

$$\text{Thus } \oint \vec{E} \cdot \hat{n} dA = \int -E_1 dA + \int E_2 \cos(70.5) dA$$

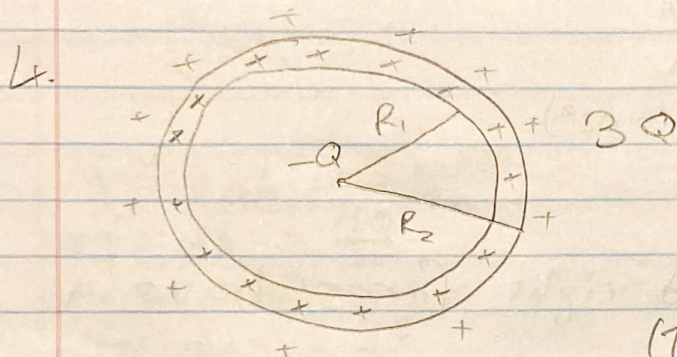
$$= -E_1 A_1 + E_2 \cos(70.5) A_2$$

$$= -(100)(0.05)(0.04) + 300 \cos(70.5)(0.05)(0.12)$$

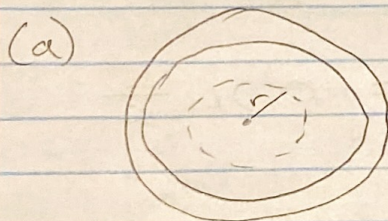
$$= 0.4 \frac{\text{N} \cdot \text{m}^2}{\text{C}}$$

So $0.4 = \frac{\Sigma q_{\text{inside}}}{\epsilon_0}$ (by Gauss' Law)

So $\Sigma q_{\text{inside}} = 3.54 \times 10^{-12} \text{ C}$



(The dotted line represents the Gaussian surface)



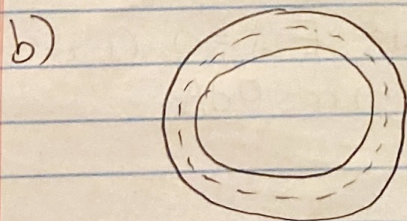
$$\oint \vec{E} \cdot \hat{n} dA = \frac{\Sigma q_{\text{inside}}}{\epsilon_0}$$

$$\text{So } \oint E dA = \frac{-Q}{\epsilon_0}$$

$$\Rightarrow -EA = \frac{-Q}{\epsilon_0}$$

$$\Rightarrow E(4\pi r^2) = \frac{Q}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q}{r^2}$$



Now $\Sigma q_{\text{inside}} = -Q + q_{\text{outer}}$ and $\frac{q_{\text{outer}}}{q_{\text{total, out}}} = \frac{V_{\text{outer}}}{V_{\text{total, out}}}$

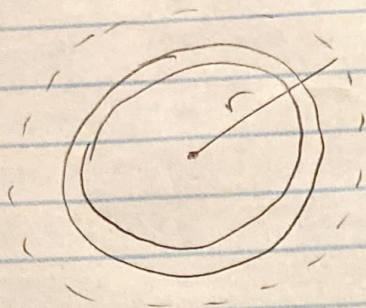
$$\Rightarrow \frac{q_{\text{outer}}}{3Q} = \frac{\frac{4}{3}\pi(r_2^3 - r_1^3)}{\frac{4}{3}\pi(R_2^3 - R_1^3)} \Rightarrow q_{\text{outer}} = \frac{3Q(r_2^3 - R_1^3)}{R_2^3 - R_1^3}$$

Thus $\oint \vec{E} \cdot \hat{n} dA = \frac{\Sigma q_{\text{ins}}}{\epsilon_0}$

$$\Rightarrow \oint E dA = \frac{-Q + 3Q \cdot \frac{r_2^3 - R_1^3}{R_2^3 - R_1^3}}{\epsilon_0}$$

$$\Rightarrow E(4\pi r^2) = \frac{-Q + 3Q \cdot \frac{r_2^3 - R_1^3}{R_2^3 - R_1^3}}{\epsilon_0} \Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{-Q + 3Q \cdot \frac{r_2^3 - R_1^3}{R_2^3 - R_1^3}}{r^2}$$

c)



$$\oint \vec{E} \cdot \hat{n} dA = \frac{\Sigma q_{\text{inside}}}{\epsilon_0}$$

$$\Rightarrow \int E dA = \frac{2Q}{\epsilon_0}$$

$$\Rightarrow EA = \frac{2Q}{\epsilon_0}$$

$$\Rightarrow E(4\pi r^2) = \frac{2Q}{\epsilon_0}$$

$$\Rightarrow E = \frac{1}{4\pi\epsilon_0} \cdot \frac{2Q}{r^2}$$

5. We have that

$$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \Sigma I_{\text{inside path}}$$

The left and right sides have $\vec{B} \perp d\vec{\ell}$, so $\vec{B} \cdot d\vec{\ell} = 0$

So $\oint \vec{B} \cdot d\vec{\ell} = -B_z L + B_x L$ (counterclockwise path)

$$= 0.6 \times 10^{-4} (0.2) + 1.5 \times 10^{-4} (0.2)$$

$$= +1.8 \times 10^{-5} \quad (1)$$

$$\text{But } \oint \vec{B} \cdot d\vec{\ell} = \mu_0 \Sigma I_{\text{inside}}$$

$$\text{So } \Sigma I_{\text{inside}} = +14.3 \text{ A}$$

Since this is positive, the current is out of the page (right hand rule)

b. We use: $\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \Sigma I_{\text{inside}}$

Taking the anti-clockwise path:

$$\oint \vec{B} \cdot d\vec{\ell} = B_{\text{bot}} L \cos(180-35) + B_{\text{top}} L \cos(180-35)$$

since the left and right sides have $\vec{B} \cdot d\vec{\ell} = 0$

$$\text{So } \oint \vec{B} \cdot d\vec{\ell} = 2(1.6 \times 10^{-4})(0.6) \cos 145$$

$$= -1.57 \times 10^{-4}$$

$$\text{So } -1.57 \times 10^{-4} = \mu_0 \Sigma I_{\text{inside}} \Rightarrow \Sigma I_{\text{in}} = -125 \text{ A}$$

The negative indicates the current goes into the page. ($-\hat{k}$) with magnitude 125 A

7.



Considering the clockwise path through the plastic ^{and}

$\oint \vec{B} \cdot d\vec{\ell} = \mu_0 \sum I_{\text{ins}}$, since B is constant and B is in the same direction as $d\ell$

(using right hand rule),

$$\oint \vec{B} \cdot d\vec{\ell} = B \oint d\ell = B L = B(2R + \pi R)$$

$$\text{Now } B(2R + \pi R) = \mu_0 \sum I_{\text{ins}}$$

Due to the N turns, $\sum I_{\text{ins}} = NI$

So $B(2R + \pi R) = \mu_0 NI$, giving:

$$B = \frac{\mu_0 NI}{2R + \pi R}$$